INCH-POUND
MIL-DTL-16377/25B(SH)
26 May 2023
SUPERSEDING
MIL-F-16377/25A(SH)
5 June 1981

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT AND SOLID-STATE EQUIVALENT, GENERAL LIGHTING, 110 WATTS, 120 VOLTS, EXPLOSIONPROOF, SYMBOLS 48.2, 68.2, 48.2SSL, AND 68.2SSL

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

Unless otherwise noted, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 13 provide information for the light fixtures covered by this specification sheet. See the requirements sections for specific references to these figures.

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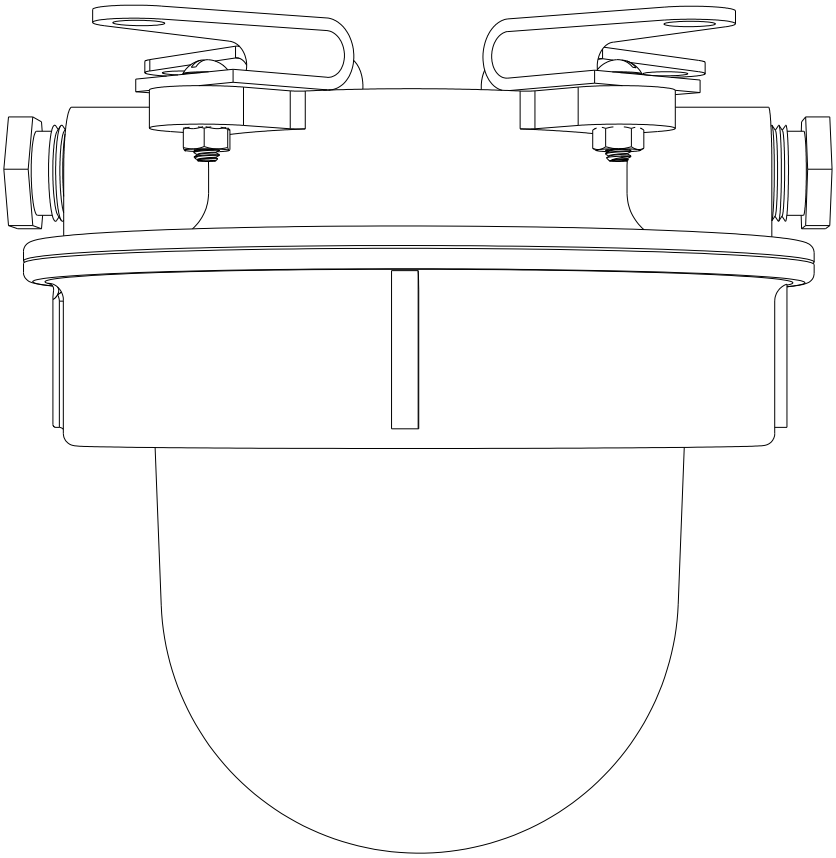


FIGURE 1. Typical view of symbols 48.2 and 68.2.

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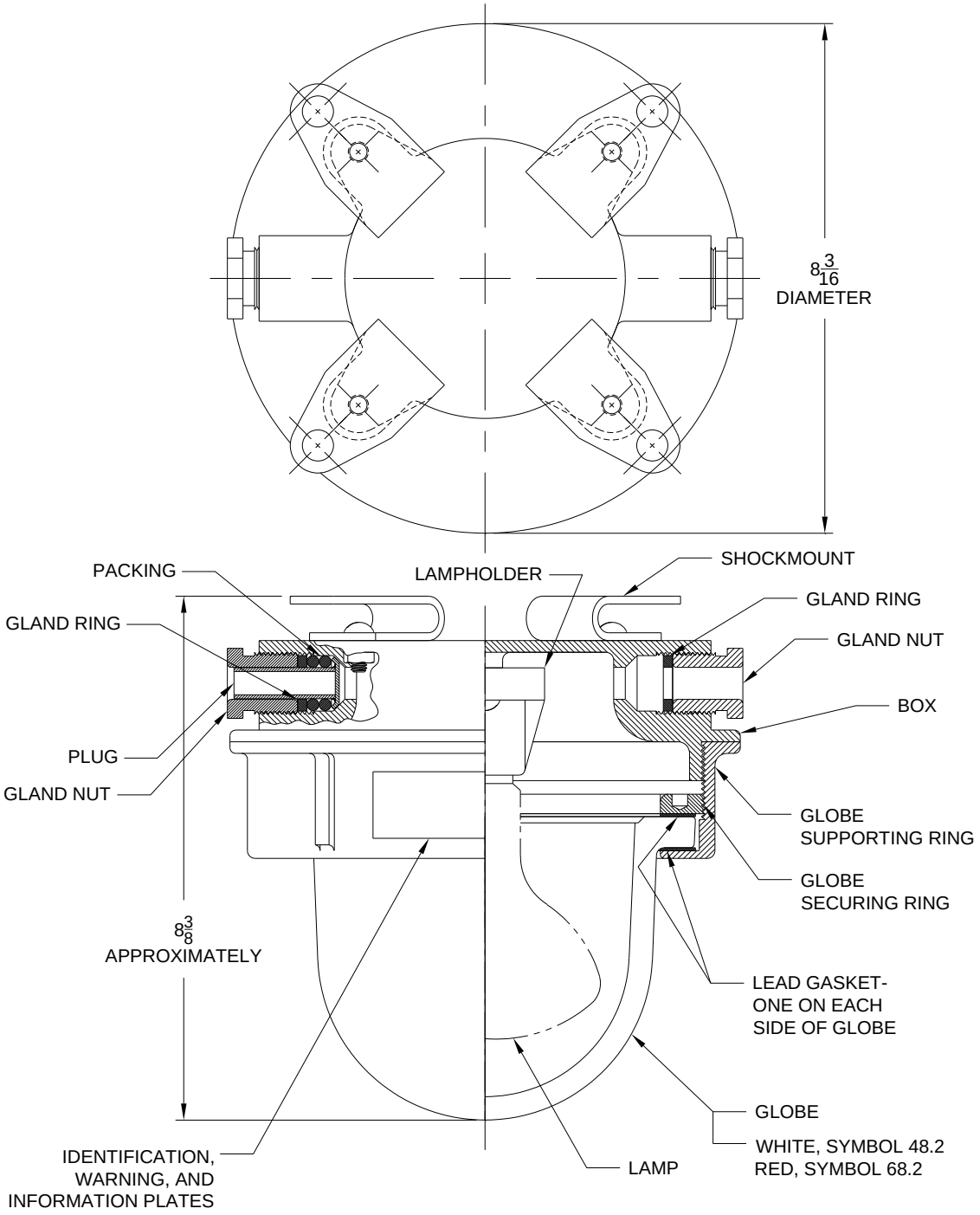


FIGURE 2. Fixture assembly.

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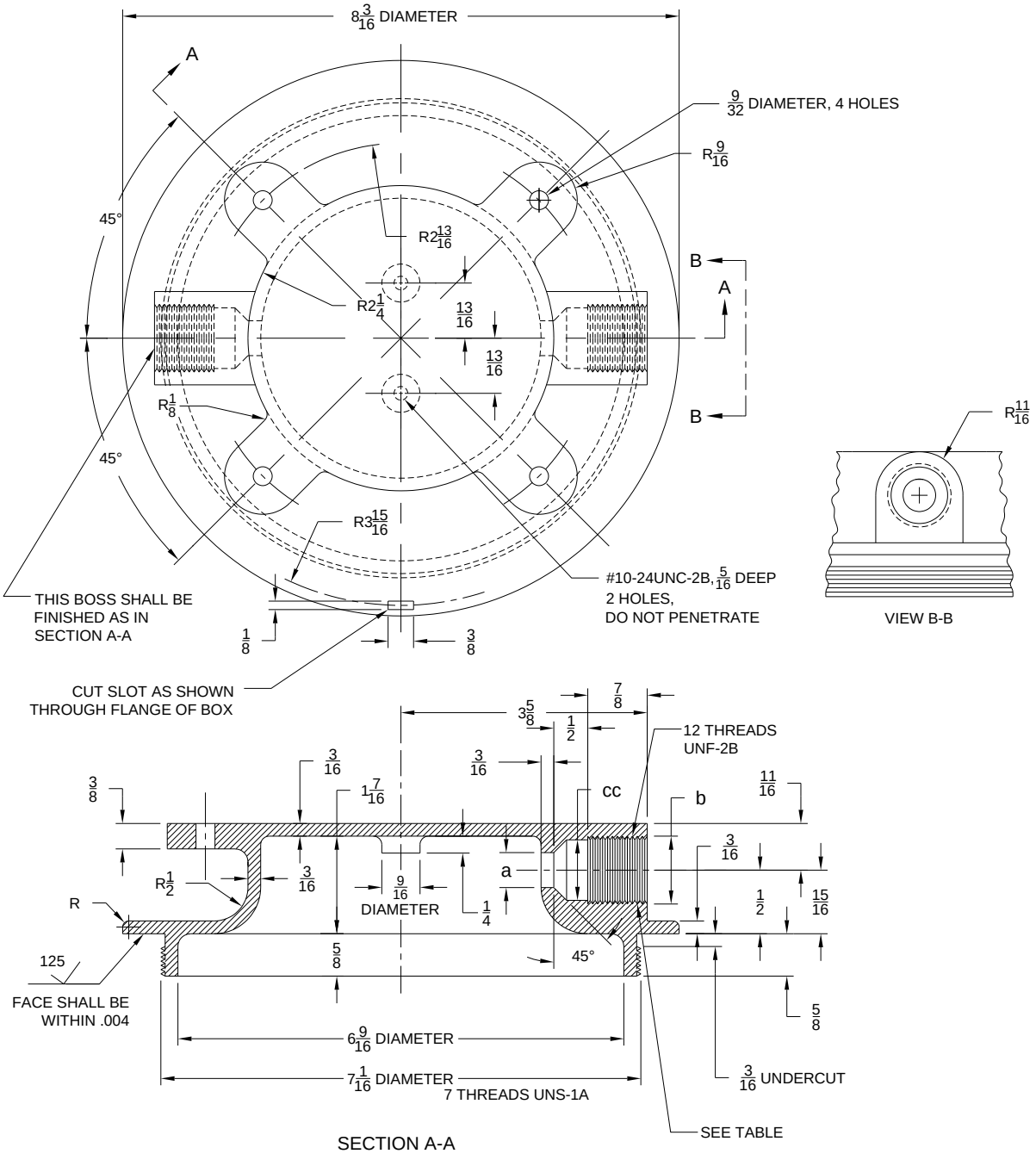


TABLE -GLAND DIMENSIONS			
SIZE	a	b	cc
"B"	.515	1	.892

FIGURE 3. Box.

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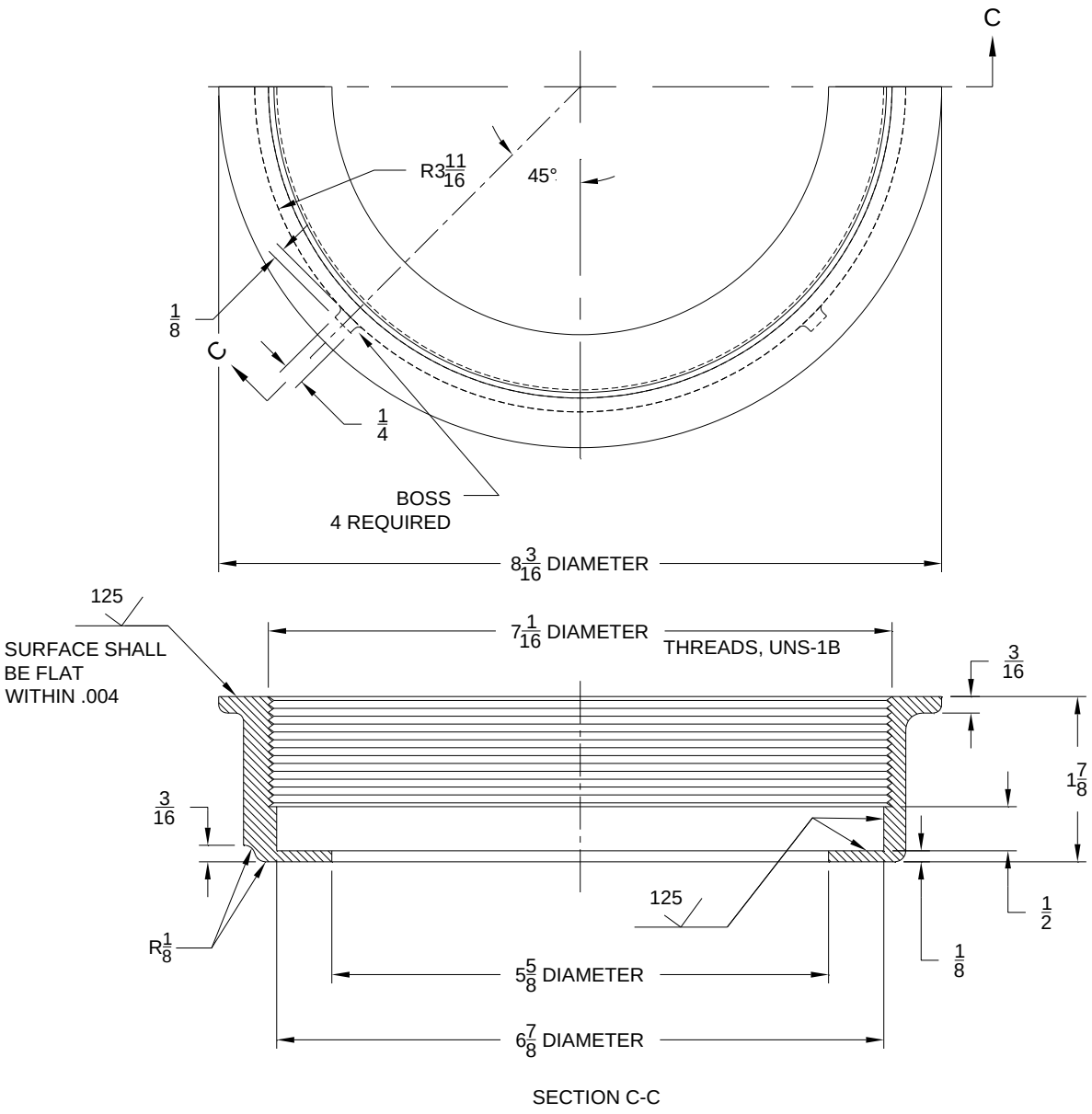


FIGURE 4. Globe supporting ring.

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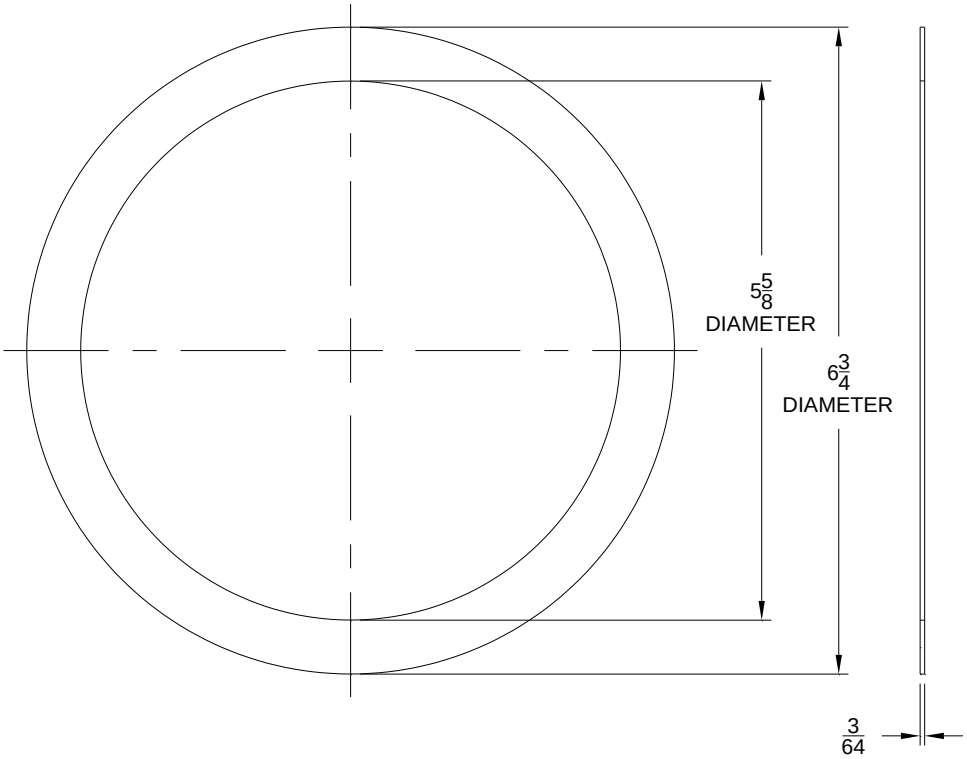


FIGURE 5. Lead gasket.

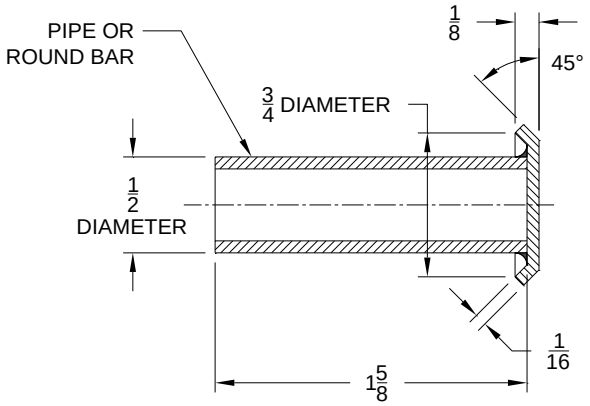


FIGURE 6. Sealing plug.

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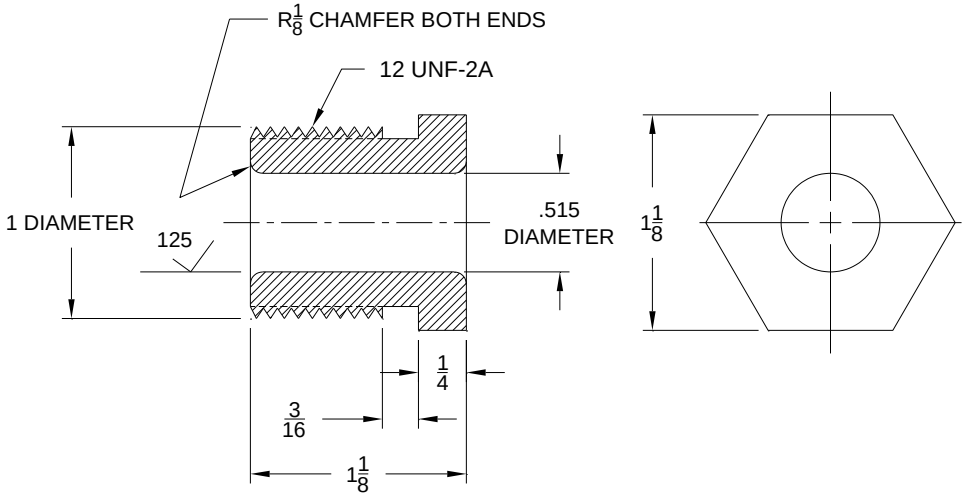


FIGURE 7. Gland nut.

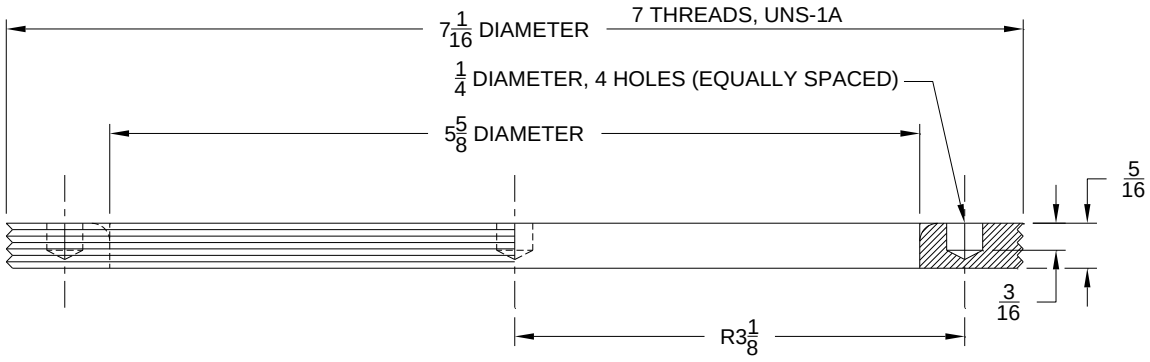
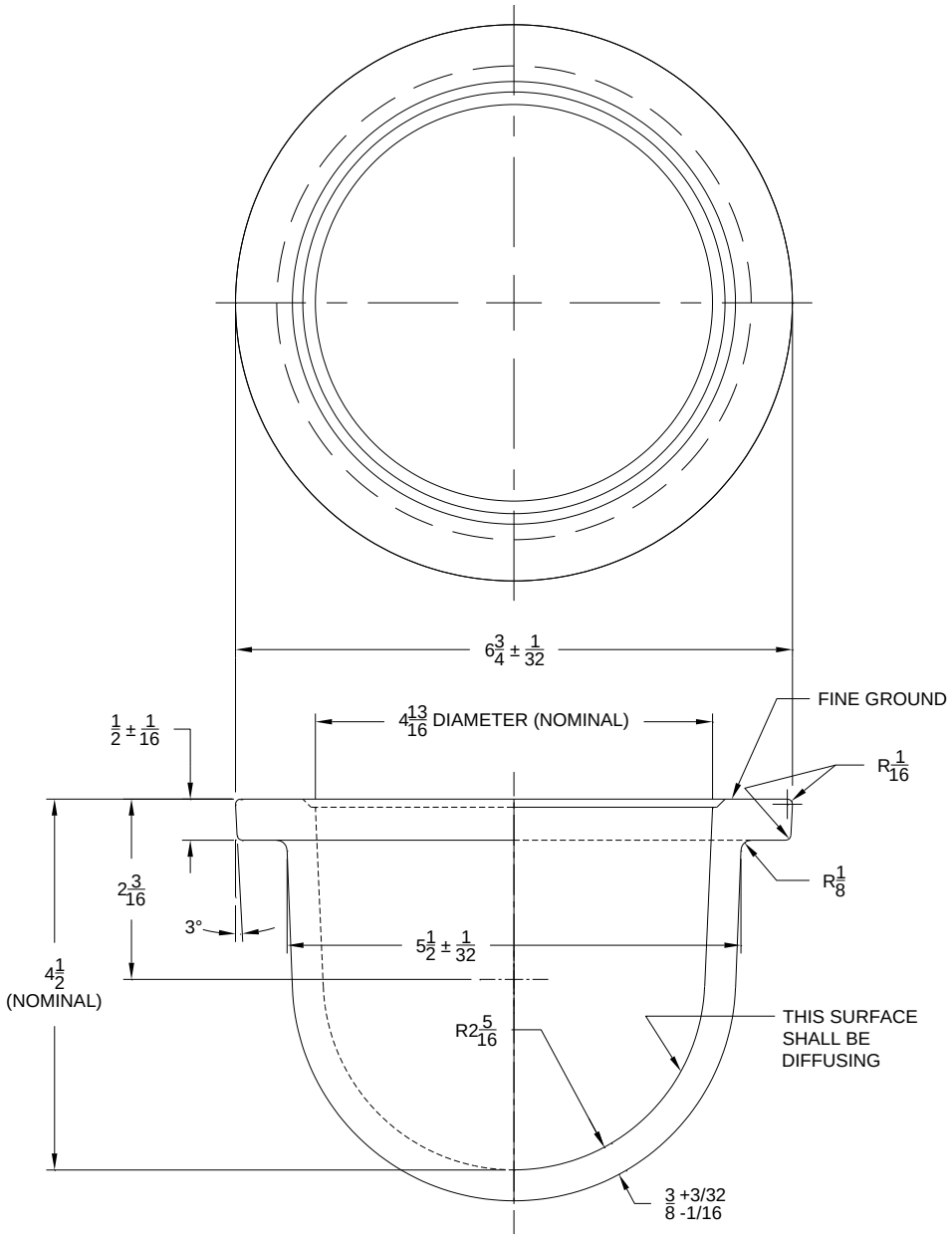


FIGURE 8. Globe securing ring.

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Material: Heat resistant, high-impact glass

Color:

Symbol 48.2: Clear transparent/diffusing

Symbol 68.2: Red transparent/diffusing

Transmittance: Symbol 48.2

Clear transparent/diffusing: 80 percent minimum

FIGURE 9. Globe.

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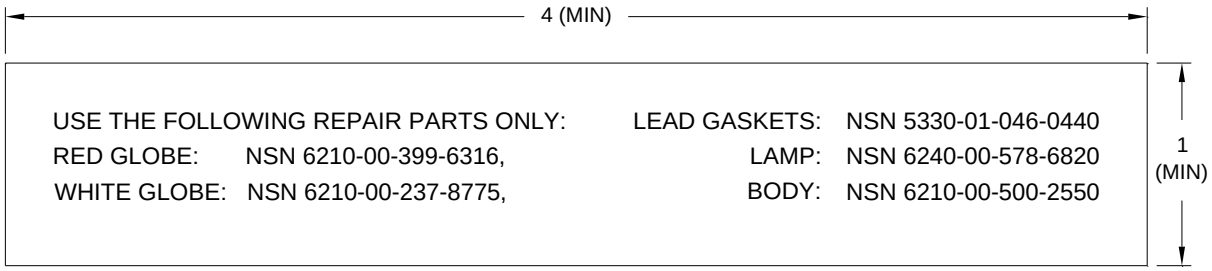


FIGURE 10. Information plate.

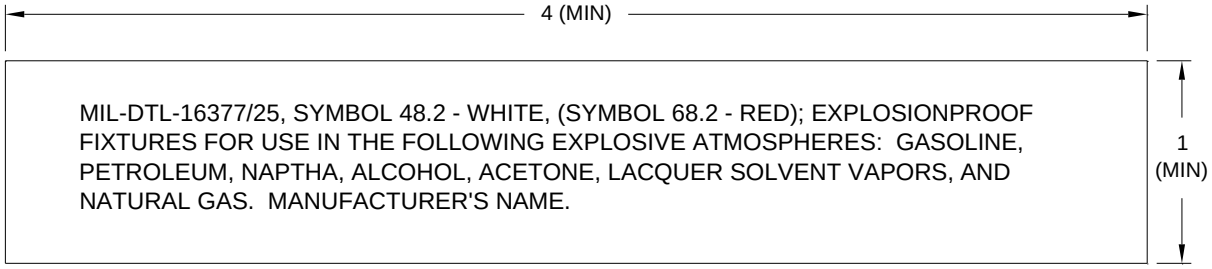


FIGURE 11. Identification plate.

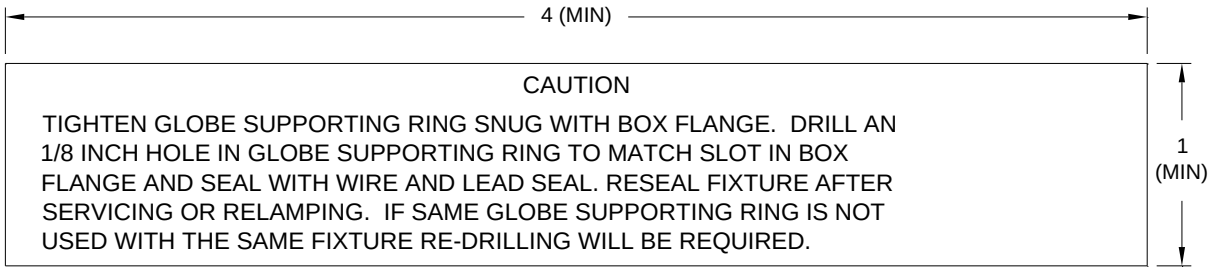
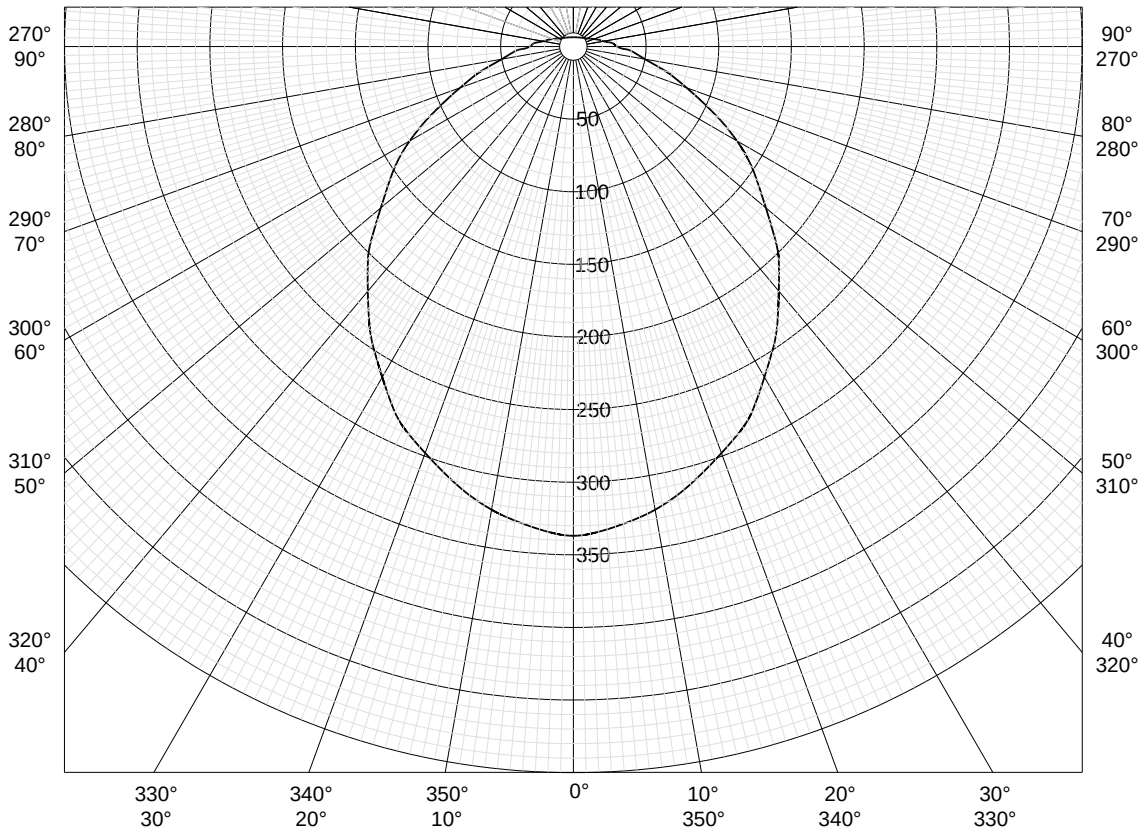


FIGURE 12. Caution plate.

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ANGLE	CANDLEPOWER SYMBOL 48.2
0	337
5	332
10	324
15	312
20	298
25	284
30	263
35	243
40	220
45	199
50	173
55	152
60	130

ANGLE	CANDLEPOWER SYMBOL 48.2
65	105
70	83
75	66
80	50
85	41
90	30
95	27
100	21
105	18
110	15
115	13
120	11
125	10

Photometric test data:

Distance: 10 feet.

Voltage: 120 volts.

Ambient temperature: 76 °F.

Lamp: 110 watts, rough service, furnished by laboratory for test, shall be used.

FIGURE 13. Candlepower distribution curves for symbol 48.2.

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INCANDESCENT REQUIREMENTS

CLASSIFICATION

Type: II
Class: 2

SYMBOL NUMBERS

48.2: Complete lighting fixture assembly with white globe.
68.2: Complete lighting fixture assembly with red globe.

DESIGN

Electrical rating:	110 watts, 120 volts.
Dimensions and configuration:	See figures 1 through 12.
Photometric data:	See figure 13.
Globes:	
Symbol 48.2:	White, see figure 9.
Symbol 68.2:	Red, see figure 9.
Enclosure:	Explosionproof.
Gland nut:	Aluminum, see figure 7.
Sealing plug:	Aluminum, see figure 6.
Gland ring:	Aluminum, ⅞ outside diameter by 0.515 inside diameter by ⅛ thick.
Box including supporting ring and globe securing ring:	Aluminum, cast.
Gaskets:	Lead, see figure 5.
Lampholder:	UL 496, Supplement SB, double-contact, medium-screw.
Shock mounts:	MIL-F-16377/41, part number M16377/41-006.
Weight:	7 pounds, 10 ounces maximum.
Lamp:	These fixtures shall be designed to accommodate a 110-watt, rough service, medium screw lamp. (Lamp not furnished.)
Designation and marking:	The following plates shall be furnished in accordance with MIL-DTL-16377 and applied as shown on figure 2. Plates shall be permanently attached to the globe supporting ring. (a) One information plate, figure 10. (b) One identification plate, figure 11. (c) One caution plate, figure 12.

SOLID-STATE LIGHTING (SSL) REQUIREMENTS

CLASSIFICATION

Type: III
Class: 2

SYMBOL NUMBER:

48.2SSL: Complete SSL lighting fixture assembly, white.
68.2SSL: Complete SSL lighting fixture assembly, red.

DESIGN

Electrical rating:	55 watts maximum, 120 volts, 60 hertz.
Dimensions and configuration:	See figures 2 through 8.

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Photometric data:	See figure 13 and 3.7.1 of MIL-DTL-16377.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.6.17 of MIL-DTL-16377.
Wiring:	The electrical connection devices chosen to terminate incoming power for each circuit (see 3.6.11.1 of MIL-DTL-16377) shall be sized to allow for both an input and output cable so that the fixtures can be installed in series within a space without the use of external junction boxes.
Enclosure:	Explosionproof.
Weight:	7 pounds, 10 ounces maximum.
Designation and marking:	One plate with the following information shall be furnished in accordance with MIL-DTL-16377 and applied as shown on figure 2: "EXPLOSIONPROOF FIXTURES FOR USE IN THE FOLLOWING EXPLOSIVE ATMOSPHERES: GASOLINE, PETROLEUM NAPHTHA, ALCOHOL, ACETONE, LACQUER SOLVENT VAPORS, AND NATURAL GAS."

EXAMINATIONS AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:	
Complete fixtures:	See table I for type I and table III for type III.
Repair parts:	See table II.
Conformance inspection:	
Comparison inspection:	Same as the first article inspection.
Conformance inspection:	See MIL-DTL-16377.

GENERAL INFORMATION

a. Assignment of military part numbers and general information is shown in table IV.

b. Rope packing in accordance with MIL-DTL-16685, class 1, composition A, compressed to ½ minimum shall be used by installing activities for cable gland sealing. DOD-HDBK-289 provides additional installation information.

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TABLE I. First article and comparison inspection for complete type II fixtures.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Shock	3.5.12	4.8.7	---
Vibration	3.5.13	4.8.8	---
Enclosure effectiveness	3.5.11	4.8.15	Explosionproof
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Continuity of grounding	3.6.13.4	4.8.12	---
Optical uniformity	3.7.5.1	4.8.4	---
Mechanical shock	3.7.5.6	4.8.17.3	8 foot-pounds
Thermal shock	3.7.5.7	4.8.17.4	---
Breakage pattern	3.7.5.9	4.8.17.6	---
Color and luminous transmittance	3.7.3.1	4.8.6.3	Symbol 68.2 only
Light output	3.7.1	4.8.6.1	Candlepower Symbol 48.2 only
Magnetic permeability	3.4.8	4.8.18	---

TABLE II. First article and comparison inspection for repair parts.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.13	4.5	---
Optical uniformity	3.7.5.1	4.8.4	---
Shock	3.5.12	4.8.7	---
Vibration	3.5.13	4.8.8	---
Light output	3.7.1	4.8.6.1	Candlepower Symbol 48.2 only
Mechanical shock	3.7.5.6	4.8.17.3	8 foot-pounds
Thermal shock	3.7.5.7	4.8.17.4	---
Breakage pattern	3.7.5.9	4.8.17.6	---
Color and luminous transmittance	3.7.3.1	4.8.6.3	Symbol 68.2 only

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TABLE III. First article and comparison inspection for complete type III fixtures.[1/]

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Power interface	3.6.18	4.8.21	---
Electromagnetic interference	3.6.14	4.8.16	---
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Shock	3.5.12	4.8.7	1/
Vibration	3.5.13	4.8.8	1/
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Noise	3.5.14	4.8.9	1/
Continuity of grounding	3.6.13.4	4.8.12	1/
Enclosure effectiveness	3.5.11	4.8.15	Explosionproof
Photometrics	3.7.1	4.8.6	---
Brightness	3.7.5	4.8.6.4	---
Chromaticity	3.7.4	4.8.6.5	---
Correlated color temperature, white light only	3.7.4.1.1	4.8.6.6	---
Lumen maintenance	3.7.11	4.8.20	---
Fail-safe circuit design	3.6.17	4.8.22	---
Optical uniformity	3.7.6.1.1.1	4.8.4	---
Stress relief	3.8.5	4.8.5	---
Magnetic permeability	3.4.8	4.8.18	---

NOTE:

1/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

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TABLE IV. Part numbers and general information.

Military Part Number M16377/25-	Item Name	National Stock Number (NSN)	Remarks
48.2	Complete fixture, symbol 48.2 Consisting of: Body White globe	 6210-00-500-2550 6210-00-237-8775	Figure 2
68.2	Complete fixture, symbol 68.2 Consisting of: Body Red transparent diffusing globe	 6210-00-500-2550 6210-00-399-6313	Figure 2
48.2SSL	Complete fixture, solid state, symbol 48.2SSL	---	---
68.2SSL	Complete fixture, solid state, symbol 68.2SSL	---	---
001	Body – complete fixtureless globe	6210-00-500-2550	Figure 2
002	White globe	---	---
003	Red transparent diffusing globe	6210-00-399-6316	---
004	Lead gaskets – a set of two gaskets	5330-01-046-0440	Figure 5
005	Globe securing ring	---	Figure 8
006	Globe supporting ring	---	Figure 4

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6210-2023-009)

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